

LeetCode SQL Problem Solving Questions With Solutions

LeetCode SQL Solutions

175. Combine Two Tables | Easy | [LeetCode](#)

Table: Person

TEXT

+-----+-----+		
Column Name	Type	
+-----+-----+		
PersonId	int	
FirstName	varchar	
LastName	varchar	
+-----+-----+		

PersonId is the primary key column for this table.

Table: Address

TEXT

+-----+	
Column Name	Type
+-----+	
AddressId	int
PersonId	int
City	varchar
State	varchar
+-----+	

AddressId is the primary key column for this table.

Write a SQL query for a report that provides the following information for each person in the Person table, regardless if there is an address for each of those people:

TEXT

FirstName, LastName, City, State

Solution

SQL

```
SELECT p.FirstName, p.LastName, a.City, a.State
FROM Person p
LEFT JOIN Address a
ON p.PersonId = a.PersonId;
```



176. Second Highest Salary | Easy | [LeetCode](#)

Write a SQL query to get the second highest salary from the Employee table.

TEXT

```
+-----+-----+
| Id | Salary |
+-----+-----+
| 1  | 100    |
| 2  | 200    |
| 3  | 300    |
+-----+-----+
```

For example, given the above Employee table, the query should return 200 as the second highest salary. If there is no second highest salary, then the query should return null.

TEXT

```
+-----+
| SecondHighestSalary |
+-----+
| 200                  |
+-----+
```

Solution

SQL

#Solution 1:

```
SELECT Max(Salary) SecondHighestSalary
FROM Employee WHERE Salary < (SELECT MAX(Salary) FROM Employee)
```

#Solution 2:

```
WITH CTE AS (SELECT DISTINCT Salary
FROM Employee
ORDER BY Salary DESC
LIMIT 2)
```

```
SELECT Salary as SecondHighestSalary
FROM CTE
ORDER BY Salary Asc
LIMIT 1;
```



```
#Solution 3:
WITH CTE AS
(
    SELECT Salary,
           DENSE_RANK() OVER (ORDER BY Salary DESC) AS DENSERANK
    FROM Employee
)
SELECT Salary SecondHighestSalary
FROM CTE
WHERE DENSERANK = 2;
```

177. Nth Highest Salary | Medium | [LeetCode](#)

Write a SQL query to get the nth highest salary from the Employee table.

```
TEXT
+-----+-----+
| Id | Salary |
+-----+-----+
| 1  | 100    |
| 2  | 200    |
| 3  | 300    |
+-----+-----+
```

For example, given the above Employee table, the nth highest salary where n = 2 is 200. If there is no nth highest salary, then the query should return null.

```
TEXT
+-----+
| getNthHighestSalary(2) |
+-----+
| 200                     |
+-----+
```


Solution

SQL



```
CREATE FUNCTION getNthHighestSalary(N INT) RETURNS INT
BEGIN
  SET N = N-1;
  RETURN(
    SELECT DISTINCT Salary FROM Employee ORDER BY Salary DESC
    LIMIT 1 OFFSET N
  );
END
```

178. Rank Scores | Medium | [LeetCode](#)

Write a SQL query to rank scores. If there is a tie between two scores, both should have the same ranking. Note that after a tie, the next ranking number should be the next consecutive integer value. In other words, there should be no “holes” between ranks.

TEXT

```
+-----+-----+
| Id | Score |
+-----+-----+
| 1 | 3.50 |
| 2 | 3.65 |
| 3 | 4.00 |
| 4 | 3.85 |
| 5 | 4.00 |
| 6 | 3.65 |
+-----+-----+
```

For example, given the above scores table, your query should generate the following report (order by highest score):

TEXT

score	Rank
4.00	1
4.00	1
3.85	2
3.65	3
3.65	3
3.50	4

Important Note: For MySQL solutions, to escape reserved words used as column names, you can use an apostrophe before and after the keyword. For example Rank .

Solution

SQL

```
SELECT score, DENSE_RANK() OVER (ORDER By Score DESC) AS "Rank"
FROM Scores;
```



180. Consecutive Numbers | Medium | [LeetCode](#)

Table: Logs

TEXT

Column Name	Type
id	int
num	varchar

id is the primary key for this table.

Write an SQL query to find all numbers that appear at least three times consecutively.

Return the result table in any order.

The query result format is in the following example:

TEXT

Logs table:

+-----+-----+	
Id Num	
+-----+-----+	
1 1	
2 1	
3 1	
4 2	
5 1	
6 2	
7 2	
+-----+-----+	

Result table:

+-----+	
ConsecutiveNums	
+-----+	
1	
+-----+	

1 is the only number that appears consecutively for at least three times.

Solution

SQL

```
SELECT a.Num as ConsecutiveNums
FROM Logs a
JOIN Logs b
ON a.id = b.id+1 AND a.num = b.num
JOIN Logs c
ON a.id = c.id+2 AND a.num = c.num;
```



181. Employees Earning More Than Their Managers | Easy | [LeetCode](#)

The `Employee` table holds all employees including their managers. Every employee has an `Id`, and there is also a column for the manager `Id`.

TEXT

Id	Name	Salary	ManagerId
1	Joe	70000	3
2	Henry	80000	4
3	Sam	60000	NULL
4	Max	90000	NULL

Given the `Employee` table, write a SQL query that finds out employees who earn more than their managers. For the above table, Joe is the only employee who earns more than his manager.

TEXT

Employee
Joe

Solution

SQL

```
SELECT E.Name as "Employee"
FROM Employee E
JOIN Employee M
```



```
ON E.ManagerId = M.Id
AND E.Salary > M.Salary;
```

182. Duplicate Emails | Easy | [LeetCode](#)

Write a SQL query to find all duplicate emails in a table named Person .

```
TEXT
+-----+-----+
| Id | Email |
+-----+-----+
| 1 | a@b.com |
| 2 | c@d.com |
| 3 | a@b.com |
+-----+-----+
```

For example, your query should return the following for the above table:

```
TEXT
+-----+
| Email |
+-----+
| a@b.com |
+-----+
```

Note: All emails are in lowercase.

Solution

```
SQL
#Solution- 1:
SELECT Email
FROM Person
GROUP BY Email
HAVING count(*) > 1
```



```
#Solution- 2:

WITH CTE AS(
SELECT Email, ROW_NUMBER() OVER(PARTITION BY Email ORDER BY Email) AS RN
      FROM Person
)

SELECT Email
FROM CTE
WHERE RN > 1;
```

183. Customers Who Never Order | Easy | [LeetCode](#)

Suppose that a website contains two tables, the `Customers` table and the `Orders` table. Write a SQL query to find all customers who never order anything.

Table: `Customers` .

TEXT

```
+-----+-----+
| Id | Name |
+-----+-----+
| 1 | Joe  |
| 2 | Henry|
| 3 | Sam  |
| 4 | Max  |
+-----+-----+
```

Table: `Orders` .

TEXT

```
+-----+-----+
| Id | CustomerId |
+-----+-----+
| 1 | 3          |
| 2 | 1          |
+-----+-----+
```

Using the above tables as example, return the following:

TEXT

```
+-----+
| Customers |
+-----+
| Henry      |
| Max        |
+-----+
```

Solution

SQL

#Solution- 1:

```
SELECT Name AS Customers
FROM Customers
LEFT JOIN Orders
ON Customers.Id = Orders.CustomerId
WHERE CustomerId IS NULL;
```

#Solution- 2:

```
SELECT Name as Customers
FROM Customers
WHERE Id NOT IN(
    SELECT CustomerId
    FROM Orders
)
```

184. Department Highest Salary | Medium | [LeetCode](#)

The `Employee` table holds all employees. Every employee has an `Id`, a salary, and there is also a column for the department `Id`.

TEXT

Id	Name	Salary	DepartmentId
1	Joe	70000	1
2	Jim	90000	1
3	Henry	80000	2
4	Sam	60000	2
5	Max	90000	1

The `Department` table holds all departments of the company.

TEXT

Id	Name
1	IT
2	Sales

Write a SQL query to find employees who have the highest salary in each of the departments. For the above tables, your SQL query should return the following rows (order of rows does not matter).

TEXT

Department	Employee	Salary
IT	Max	90000
IT	Jim	90000
Sales	Henry	80000

Explanation:

Max and Jim both have the highest salary in the IT department and Henry has the

highest salary in the Sales department.

Solution

SQL



```
SELECT Department.Name AS Department, Employee.Name AS Employee, Salary
FROM Employee
JOIN Department
ON Employee.DepartmentId = Department.Id
WHERE (DepartmentId, Salary) IN(
    SELECT DepartmentId, MAX(Salary) AS Salary
    FROM Employee
    GROUP BY DepartmentId
);
```

185. Department Top Three Salaries | Hard | [LeetCode](#)

The `Employee` table holds all employees. Every employee has an `Id`, and there is also a column for the department `Id`.

TEXT

```
+----+-----+-----+-----+
| Id | Name` | Salary | DepartmentId |
+----+-----+-----+-----+
| 1  | Joe   | 85000  | 1             |
| 2  | Henry | 80000  | 2             |
| 3  | Sam   | 60000  | 2             |
| 4  | Max   | 90000  | 1             |
| 5  | Janet | 69000  | 1             |
| 6  | Randy | 85000  | 1             |
| 7  | Will  | 70000  | 1             |
+----+-----+-----+-----+
```

The `Department` table holds all departments of the company.

TEXT

+-----+-----+		
Id	Name	
+-----+-----+		
1	IT	
2	Sales	
+-----+-----+		

Write a SQL query to find employees who earn the top three salaries in each of the department. For the above tables, your SQL query should return the following rows (order of rows does not matter).

TEXT

+-----+-----+-----+			
Department	Employee	Salary	
+-----+-----+-----+			
IT	Max	90000	
IT	Randy	85000	
IT	Joe	85000	
IT	Will	70000	
Sales	Henry	80000	
Sales	Sam	60000	
+-----+-----+-----+			

Explanation:

In IT department, Max earns the highest salary, both Randy and Joe earn the second highest salary, and Will earns the third highest salary. There are only two employees in the Sales department, Henry earns the highest salary while Sam earns the second highest salary.

Solution

SQL

```
WITH department_ranking AS (  
  SELECT Name AS Employee, Salary, DepartmentId  
         ,DENSE_RANK() OVER (PARTITION BY DepartmentId ORDER BY Salary DESC) AS rnk  
  FROM Employee
```



)

```
SELECT d.Name AS Department, r.Employee, r.Salary
FROM department_ranking AS r
JOIN Department AS d
ON r.DepartmentId = d.Id
WHERE r.rnk <= 3
ORDER BY d.Name ASC, r.Salary DESC;
```

196. Delete Duplicate Emails | Easy | [LeetCode](#)

Write a SQL query to delete all duplicate email entries in a table named `Person`, keeping only unique emails based on its smallest `Id`.

TEXT

Id	Email
1	john@example.com
2	bob@example.com
3	john@example.com

`Id` is the primary key column for this table. For example, after running your query, the above `Person` table should have the following rows:

TEXT

Id	Email
1	john@example.com
2	bob@example.com

Note:

Your output is the whole `Person` table after executing your sql. Use `delete`

statement.

Solution

SQL

```
DELETE p2
FROM Person p1
JOIN Person p2
ON p1.Email = p2.Email
AND p1.id < p2.id
```



197. Rising Temperature | Easy | [LeetCode](#)

Table: Weather

TEXT

+-----+-----+		
Column Name	Type	
+-----+-----+		
id	int	
recordDate	date	
temperature	int	
+-----+-----+		

id is the primary key for this table.

This table contains information about the temperature in a certain day.

Write an SQL query to find all dates' id with higher temperature compared to its previous dates (yesterday).

Return the result table in any order.

The query result format is in the following example:

TEXT

Weather

+-----+-----+-----+-----+			
id	recordDate	Temperature	
+-----+-----+-----+-----+			
1	2015-01-01	10	
2	2015-01-02	25	
3	2015-01-03	20	
4	2015-01-04	30	
+-----+-----+-----+-----+			

Result table:

+-----+	
id	
+-----+	
2	
4	
+-----+	

In 2015-01-02, temperature was higher than the previous day (10 -> 25).

In 2015-01-04, temperature was higher than the previous day (20 -> 30).

Solution

SQL

#Solution- 1:

```
SELECT t.Id
FROM Weather AS t, Weather AS y
WHERE DATEDIFF(t.RecordDate, y.RecordDate) = 1
AND t.Temperature > y.Temperature;
```

#Solution- 2:

```
SELECT t.Id
FROM Weather t
JOIN Weather y
ON DATEDIFF(t.recordDate, y.recordDate) = 1 AND
t.temperature > y.temperature;
```



262. Trips and Users | Hard | [LeetCode](#)

Table: Trips

TEXT

Column Name	Type	
Id	int	
Client_Id	int	
Driver_Id	int	
City_Id	int	
Status	enum	
Request_at	date	

Id is the primary key for this table.

The table holds all taxi trips. Each trip has a unique Id, while Client_Id and Driver_Id are foreign keys to the Users table (Users_Id) referring to the user that either made the trip or accepted the trip. Status is an ENUM type of ('completed', 'cancelled_by_driver', 'cancelled_by_client').

Table: Users

TEXT

Column Name	Type	
Users_Id	int	
Banned	enum	
Role	enum	

Users_Id is the primary key for this table.

The table holds all users. Each user has a unique Users_Id, and Role is an ENUM type of ('client', 'partner', 'driver', 'partner'). Banned is an ENUM type of ('Yes', 'No').

Write a SQL query to find the cancellation rate of requests with unbanned users (both client and driver must not be banned) each day between "2013-10-01" and "2013-10-03".

The cancellation rate is computed by dividing the number of canceled (by client or driver) requests with unbanned users by the total number of requests with unbanned users on that day.

Return the result table in any order. Round Cancellation Rate to two decimal points.

The query result format is in the following example:

TEXT

Trips table:

+-----+-----+-----+-----+-----+-----+-----+						
Id	Client_Id	Driver_Id	City_Id	Status	Request_at	
+-----+-----+-----+-----+-----+-----+-----+						
1	1	10	1	completed	2013-10-01	
2	2	11	1	cancelled_by_driver	2013-10-01	
3	3	12	6	completed	2013-10-01	
4	4	13	6	cancelled_by_client	2013-10-01	
5	1	10	1	completed	2013-10-02	
6	2	11	6	completed	2013-10-02	
7	3	12	6	completed	2013-10-02	
8	2	12	12	completed	2013-10-03	
9	3	10	12	completed	2013-10-03	
10	4	13	12	cancelled_by_driver	2013-10-03	
+-----+-----+-----+-----+-----+-----+-----+						

Users table:

+-----+-----+-----+			
Users_Id	Banned	Role	
+-----+-----+-----+			
1	No	client	
2	Yes	client	
3	No	client	
4	No	client	
10	No	driver	
11	No	driver	
12	No	driver	
13	No	driver	
+-----+-----+-----+			

Result table:

Day	Cancellation Rate
2013-10-01	0.33
2013-10-02	0.00
2013-10-03	0.50

On 2013-10-01:

- There were 4 requests in total, 2 of which were canceled.
- However, the request with Id=2 was made by a banned client (User_Id=2), so it is ignored.
- Hence there are 3 unbanned requests in total, 1 of which was canceled.
- The Cancellation Rate is $(1 / 3) = 0.33$

On 2013-10-02:

- There were 3 requests in total, 0 of which were canceled.
- The request with Id=6 was made by a banned client, so it is ignored.
- Hence there are 2 unbanned requests in total, 0 of which were canceled.
- The Cancellation Rate is $(0 / 2) = 0.00$

On 2013-10-03:

- There were 3 requests in total, 1 of which was canceled.
- The request with Id=8 was made by a banned client, so it is ignored.
- Hence there are 2 unbanned request in total, 1 of which were canceled.
- The Cancellation Rate is $(1 / 2) = 0.50$



Solution

SQL



```
SELECT Request_at AS Day,
ROUND(SUM(IF(Status<>'completed', 1, 0))/COUNT(Status),2) AS "Cancellation Rate"
FROM Trips
WHERE Request_at BETWEEN "2013-10-01" AND "2013-10-03"
AND Client_Id NOT IN (SELECT Users_Id FROM Users WHERE Banned = 'Yes')
AND Driver_Id NOT IN (SELECT Users_Id FROM Users WHERE Banned = 'Yes')
GROUP BY Request_at;
```



511. Game Play Analysis I | Easy | [LeetCode](#)

Table: Activity

TEXT

+-----+-----+		
Column Name	Type	
+-----+-----+		
player_id	int	
device_id	int	
event_date	date	
games_played	int	
+-----+-----+		

(player_id, event_date) is the primary key of this table.

This table shows the activity of players of some game.

Each row is a record of a player who logged in and played a number of games (poss

Write an SQL query that reports the **first login date** for each player.

The query result format is in the following example:

TEXT

Activity table:

+-----+-----+-----+-----+				
player_id	device_id	event_date	games_played	
+-----+-----+-----+-----+				
1	2	2016-03-01	5	
1	2	2016-05-02	6	
2	3	2017-06-25	1	
3	1	2016-03-02	0	
3	4	2018-07-03	5	
+-----+-----+-----+-----+				

Result table:

+-----+-----+		
player_id	first_login	
+-----+-----+		
1	2016-03-01	

2	2017-06-25	
3	2016-03-02	
+-----+-----+		

Solution

SQL

```
SELECT player_id, MIN(event_date) as first_login
FROM Activity
GROUP BY player_id
```



512. Game Play Analysis II | Easy | [LeetCode](#)

Table: Activity

TEXT

+-----+-----+		
Column Name	Type	
+-----+-----+		
player_id	int	
device_id	int	
event_date	date	
games_played	int	
+-----+-----+		

(player_id, event_date) is the primary key of this table.

This table shows the activity of players of some game.

Each row is a record of a player who logged in and played a number of games (poss

Write a SQL query that reports the device that is first logged in for each player.

The query result format is in the following example:

TEXT

Activity table:

player_id	device_id	event_date	games_played
1	2	2016-03-01	5
1	2	2016-05-02	6
2	3	2017-06-25	1
3	1	2016-03-02	0
3	4	2018-07-03	5

Result table:

player_id	device_id
1	2
2	3
3	1

Solution

SQL

#Solution- 1:

```
SELECT DISTINCT player_id, device_id
FROM Activity
WHERE (player_id, event_date) in (
    SELECT player_id, min(event_date)
    FROM Activity
    GROUP BY player_id)
```

#Solution- 2:

```
SELECT a.player_id, b.device_id
FROM
(SELECT player_id, MIN(event_date) AS event_date FROM Activity
GROUP BY player_id) a
JOIN Activity b
```



```
ON a.player_id = b.player_id AND a.event_date = b.event_date;
```

#Solution- 3:

```
SELECT player_id, device_id
FROM
(SELECT player_id, device_id, event_date,
ROW_NUMBER() OVER (PARTITION BY player_id ORDER BY event_date) AS r
FROM Activity) lookup
WHERE r = 1;
```

534. Game Play Analysis III | Medium | [LeetCode](#)

Table: Activity

TEXT

```
+-----+-----+
| Column Name | Type |
+-----+-----+
| player_id   | int  |
| device_id   | int  |
| event_date  | date |
| games_played | int  |
+-----+-----+
```

(player_id, event_date) is the primary key of this table.

This table shows the activity of players of some game.

Each row is a record of a player who logged in and played a number of games (poss

Write an SQL query that reports for each player and date, how many games played so far by the player. That is, the total number of games played by the player until that date. Check the example for clarity.

The query result format is in the following example:

TEXT

Activity table:

```
+-----+-----+-----+-----+
```

player_id	device_id	event_date	games_played
1	2	2016-03-01	5
1	2	2016-05-02	6
1	3	2017-06-25	1
3	1	2016-03-02	0
3	4	2018-07-03	5

Result table:

player_id	event_date	games_played_so_far
1	2016-03-01	5
1	2016-05-02	11
1	2017-06-25	12
3	2016-03-02	0
3	2018-07-03	5

For the player with id 1, $5 + 6 = 11$ games played by 2016-05-02, and $5 + 6 + 1 =$

For the player with id 3, $0 + 5 = 5$ games played by 2018-07-03.

Note that for each player we only care about the days when the player logged in.



Solution

SQL



#Solution- 1:

```
SELECT t1.player_id, t1.event_date, SUM(t2.games_played) as games_played_so_far
FROM Activity t1
JOIN Activity t2
ON t1.player_id = t2.player_id
WHERE t1.event_date >= t2.event_date
GROUP BY t1.player_id, t1.event_date;
```

#Solution- 2:

```
SELECT player_id, event_date,
```

```
SUM(games_played) OVER (PARTITION BY player_id ORDER BY event_date) AS games_play
FROM Activity;
```

550. Game Play Analysis IV | Medium | [LeetCode](#)

Table: Activity

TEXT

```
+-----+-----+
| Column Name | Type |
+-----+-----+
| player_id   | int  |
| device_id   | int  |
| event_date  | date |
| games_played | int  |
+-----+-----+
```

(player_id, event_date) is the primary key of this table.

This table shows the activity of players of some game.

Each row is a record of a player who logged in and played a number of games (poss

Write an SQL query that reports the fraction of players that logged in again on the day after the day they first logged in, rounded to 2 decimal places. In other words, you need to count the number of players that logged in for at least two consecutive days starting from their first login date, then divide that number by the total number of players.

The query result format is in the following example:

TEXT

Activity table:

```
+-----+-----+-----+-----+
| player_id | device_id | event_date | games_played |
+-----+-----+-----+-----+
| 1         | 2         | 2016-03-01 | 5             |
| 1         | 2         | 2016-03-02 | 6             |
| 2         | 3         | 2017-06-25 | 1             |
```

3	1	2016-03-02	0
3	4	2018-07-03	5

Result table:

fraction
0.33

Only the player with id 1 logged back in after the first day he had logged in so

Solution

SQL



#Solution- 1:

```
SELECT ROUND(sum(CASE WHEN t1.event_date = t2.first_event+1 THEN 1 ELSE 0 END))/COUNT(*)
FROM Activity t1
JOIN
  (SELECT player_id, MIN(event_date) AS first_event
   FROM Activity
   GROUP BY player_id) t2
ON t1.player_id = t2.player_id;
```

#Solution- 2:

```
SELECT ROUND(COUNT(DISTINCT b.player_id)/COUNT(DISTINCT a.player_id),2) AS fraction
FROM
  (SELECT player_id, MIN(event_date) AS event_date FROM Activity
   GROUP BY player_id) a
LEFT JOIN Activity b
ON a.player_id = b.player_id AND a.event_date+1 = b.event_date;
```

The `Employee` table holds all employees. The employee table has three columns: Employee Id, Company Name, and Salary.

TEXT

+-----+-----+-----+			
Id	Company	Salary	
+-----+-----+-----+			
1	A	2341	
2	A	341	
3	A	15	
4	A	15314	
5	A	451	
6	A	513	
7	B	15	
8	B	13	
9	B	1154	
10	B	1345	
11	B	1221	
12	B	234	
13	C	2345	
14	C	2645	
15	C	2645	
16	C	2652	
17	C	65	
+-----+-----+-----+			

Write a SQL query to find the median salary of each company. Bonus points if you can solve it without using any built-in SQL functions.

TEXT

+-----+-----+-----+			
Id	Company	Salary	
+-----+-----+-----+			
5	A	451	
6	A	513	
12	B	234	
9	B	1154	
14	C	2645	
+-----+-----+-----+			

Solution

SQL



```
SELECT t1.Id AS Id, t1.Company, t1.Salary
FROM Employee AS t1 JOIN Employee AS t2
ON t1.Company = t2.Company
GROUP BY t1.Id
HAVING abs(sum(CASE WHEN t2.Salary<t1.Salary THEN 1
                    WHEN t2.Salary>t1.Salary THEN -1
                    WHEN t2.Salary=t1.Salary AND t2.Id<t1.Id THEN 1
                    WHEN t2.Salary=t1.Salary AND t2.Id>t1.Id THEN -1
                    ELSE 0 END)) <= 1
ORDER BY t1.Company, t1.Salary, t1.Id
```

570. Managers with at Least 5 Direct Reports | Medium |

[LeetCode](#)

The `Employee` table holds all employees including their managers. Every employee has an `Id`, and there is also a column for the manager `Id`.

TEXT

+-----+-----+-----+-----+			
Id	Name	Department	ManagerId
+-----+-----+-----+-----+			
101	John	A	null
102	Dan	A	101
103	James	A	101
104	Amy	A	101
105	Anne	A	101
106	Ron	B	101
+-----+-----+-----+-----+			

Given the `Employee` table, write a SQL query that finds out managers with at least 5 direct report. For the above table, your SQL query should return:

TEXT

```
+-----+
| Name  |
+-----+
| John  |
+-----+
```

Note: No one would report to himself.

Solution

SQL

```
SELECT Name
FROM Employee
WHERE id IN
    (SELECT ManagerId
     FROM Employee
     GROUP BY ManagerId
     HAVING COUNT(DISTINCT Id) >= 5)
```



571. Find Median Given Frequency of Numbers | [LeetCode](https://leetcode.com/problems/find-median-given-frequency-of-numbers/)

The `Numbers` table keeps the value of number and its frequency.

TEXT

```
+-----+-----+
| Number | Frequency |
+-----+-----+
| 0      | 7         |
| 1      | 1         |
| 2      | 3         |
| 3      | 1         |
+-----+-----+
```

In this table, the numbers are 0, 0, 0, 0, 0, 0, 0, 0, 1, 2, 2, 2, 3, so the median is $(0 + 0) / 2 = 0$.

TEXT

```
+-----+
| median |
+-----+
| 0.0000 |
+-----+
```

Write a query to find the median of all numbers and name the result as median.

Solution

SQL



```
SELECT avg(t3.Number) as median
FROM Numbers as t3
JOIN
  (SELECT t1.Number,
    abs(SUM(CASE WHEN t1.Number>t2.Number THEN t2.Frequency ELSE 0 END) -
      SUM(CASE WHEN t1.Number<t2.Number THEN t2.Frequency ELSE 0 END)) AS count_diff
  FROM numbers AS t1, numbers AS t2
  GROUP BY t1.Number) AS t4
ON t3.Number = t4.Number
WHERE t3.Frequency>=t4.count_diff
```

574. Winning Candidate | Medium | [LeetCode](#)

Table: Candidate

TEXT

```
+-----+-----+
| id  | Name  |
+-----+-----+
| 1   | A     |
```

	2		B	
	3		C	
	4		D	
	5		E	
+-----+-----+				

Table: vote

TEXT

+-----+-----+				
	id		CandidateId	
+-----+-----+				
	1		2	
	2		4	
	3		3	
	4		2	
	5		5	
+-----+-----+				

id is the auto-increment primary key, CandidateId is the id appeared in Candidate table. Write a sql to find the name of the winning candidate, the above example will return the winner B.

TEXT

+-----+		
	Name	
+-----+		
	B	
+-----+		

Notes: You may assume there is no tie, in other words there will be at most one winning candidate.

Solution



SQL

```
SELECT Name
FROM Candidate
WHERE id = (SELECT CandidateId
            FROM Vote
            GROUP BY CandidateId
            ORDER BY COUNT(1) desc
            LIMIT 1)
```

```
## Assumption: if we have two candidates with the same votes, we choose the one w
# SELECT Name
# FROM Candidate JOIN
# .... (SELECT CandidateId
# .... FROM Vote
# .... GROUP BY CandidateId
# .... ORDER BY count(1) DESC
# .... LIMIT 1) AS t
# ON Candidate.id = t.CandidateId
```

577. Employee Bonus | Easy | [LeetCode](#)

Select all employee's name and bonus whose bonus is < 1000.

Table:Employee

TEXT

```
+-----+-----+-----+-----+
| empId | name  | supervisor| salary |
+-----+-----+-----+-----+
| 1     | John  | 3         | 1000   |
| 2     | Dan   | 3         | 2000   |
| 3     | Brad  | null      | 4000   |
| 4     | Thomas| 3         | 4000   |
+-----+-----+-----+-----+
empId is the primary key column for this table.
```

Table: Bonus

TEXT

```
+-----+-----+
| empId | bonus |
+-----+-----+
| 2     | 500   |
| 4     | 2000  |
+-----+-----+
```

empId is the primary key column for this table.

Example output:

TEXT

```
+-----+-----+
| name  | bonus |
+-----+-----+
| John  | null  |
| Dan   | 500   |
| Brad  | null  |
+-----+-----+
```

Solution

SQL

```
SELECT name, bonus
FROM Employee LEFT JOIN Bonus
ON Employee.empId = Bonus.empId
WHERE bonus < 1000 OR bonus IS NULL;
```



578. Get Highest Answer Rate Question | Medium | [LeetCode](https://leetcode.com/problems/get-highest-answer-rate-question/)

Get the highest answer rate question from a table *surveylog* with these columns: *uid*, *action*, *questionid*, *answerid*, *qnum*, *timestamp*.

uid means user id; action has these kind of values: "show", "answer", "skip"; *answerid is not null when action column is "answer", while is null for "show" and "skip"*; qnum is the numeral order of the question in current session.

Write a sql query to identify the question which has the highest answer rate.

Example: Input:

TEXT

uid	action	question_id	answer_id	q_num	timestamp
5	show	285	null	1	123
5	answer	285	124124	1	124
5	show	369	null	2	125
5	skip	369	null	2	126

Output:

TEXT

survey_log
285

Explanation: question 285 has answer rate 1/1, while question 369 has 0/1 answer rate, so output 285.

Note: The highest answer rate meaning is: answer number's ratio in show number in the same question.

Solution



SQL

#Solution- 1::

```
SELECT question_id AS survey_log FROM
(SELECT question_id,
        SUM(IF(action='show', 1, 0)) AS num_show,
        SUM(IF(action='answer', 1, 0)) AS num_answer
 FROM survey_log GROUP BY question_id) AS t
ORDER BY (num_answer/num_show) DESC LIMIT 1;
```

#Solution- 2:

```
SELECT question_id AS survey_log
FROM (SELECT question_id,
        sum(CASE WHEN action='show' THEN 1 ELSE 0 END) AS show_count,
        sum(CASE WHEN action='answer' THEN 1 ELSE 0 END) AS answer_count
 FROM survey_log
 GROUP BY question_id) AS t
ORDER BY answer_count/show_count DESC LIMIT 1;
```

579. Find Cumulative Salary of an Employee | Hard |

LeetCode

The Employee table holds the salary information in a year.

Write a SQL to get the cumulative sum of an employee's salary over a period of 3 months but exclude the most recent month.

The result should be displayed by 'Id' ascending, and then by 'Month' descending.

Example Input

TEXT

Id	Month	Salary
1	1	20
2	1	20
1	2	30
2	2	30
3	2	40

	1		3		40	
	3		3		60	
	1		4		60	
	3		4		70	

Output

TEXT

	Id		Month		Salary	
	----		-----		-----	
	1		3		90	
	1		2		50	
	1		1		20	
	2		1		20	
	3		3		100	
	3		2		40	

Explanation Employee '1' has 3 salary records for the following 3 months except the most recent month '4': salary 40 for month '3', 30 for month '2' and 20 for month '1' So the cumulative sum of salary of this employee over 3 months is 90(40+30+20), 50(30+20) and 20 respectively.

TEXT

	Id		Month		Salary	
	----		-----		-----	
	1		3		90	
	1		2		50	
	1		1		20	

Employee '2' only has one salary record (month '1') except its most recent month '2'.

TEXT

	Id		Month		Salary	
	----		-----		-----	
	2		1		20	

Employee '3' has two salary records except its most recent pay month '4': month '3' with 60 and month '2' with 40. So the cumulative salary is as following.

TEXT

Id	Month	Salary
3	3	100
3	2	40

Solution

SQL

```
SELECT
    a.id,
    a.month,
    SUM(b.salary) Salary
FROM
    Employee a JOIN Employee b ON
    a.id = b.id AND
    a.month - b.month >= 0 AND
    a.month - b.month < 3
GROUP BY
    a.id, a.month
HAVING
    (a.id, a.month) NOT IN (SELECT id, MAX(month) FROM Employee GROUP BY id)
ORDER BY
    a.id, a.month DESC
```

580. Count Student Number in Departments | Medium |

[LeetCode](#)

A university uses 2 data tables, `student` and `department`, to store data about its students and the departments associated with each major.

Write a query to print the respective department name and number of students majoring in each department for all departments in the department table (even ones

with no current students).

Sort your results by descending number of students; if two or more departments have the same number of students, then sort those departments alphabetically by department name.

The `student` is described as follow:

TEXT

Column Name	Type
student_id	Integer
student_name	String
gender	Character
dept_id	Integer

where *studentid* is the student's ID number, *studentname* is the student's name, *gender* is their gender, and *dept_id* is the department ID associated with their declared major.

And the department table is described as below:

TEXT

Column Name	Type
dept_id	Integer
dept_name	String

where *deptid* is the department's ID number and *deptname* is the department name.

Here is an example input: `student` table:

TEXT

student_id	student_name	gender	dept_id
1	Jack	M	1
2	Jane	F	1
3	Mark	M	2

department table:

TEXT

dept_id	dept_name
1	Engineering
2	Science
3	Law

The Output should be:

TEXT

dept_name	student_number
Engineering	2
Science	1
Law	0

Solution

SQL

```
SELECT dept_name,  
       SUM(CASE WHEN student_id IS NULL THEN 0 ELSE 1 END) AS student_number  
FROM department  
LEFT JOIN student  
ON department.dept_id = student.dept_id  
GROUP BY department.dept_id  
ORDER BY student_number DESC, dept_name
```



584. Find Customer Referee | Easy | [LeetCode](#)

Given a table `customer` holding customers information and the referee.

TEXT

id	name	referee_id
1	Will	NULL
2	Jane	NULL
3	Alex	2
4	Bill	NULL
5	Zack	1
6	Mark	2

Write a query to return the list of customers NOT referred by the person with id '2'.

For the sample data above, the result is:

TEXT

name
Will
Jane
Bill
Zack

Solution

SQL

```
SELECT name
FROM customer
WHERE referee_id != '2' OR referee_id IS NULL;
```



Write a query to print the sum of all total investment values in 2016 (TIV_2016), to a scale of 2 decimal places, for all policy holders who meet the following criteria:

1. Have the same TIV_2015 value as one or more other policyholders.
2. Are not located in the same city as any other policyholder (i.e.: the (latitude, longitude) attribute pairs must be unique). Input Format: The insurance table is described as follows:

TEXT

Column Name	Type
PID	INTEGER(11)
TIV_2015	NUMERIC(15,2)
TIV_2016	NUMERIC(15,2)
LAT	NUMERIC(5,2)
LON	NUMERIC(5,2)

where PID is the policyholder's policy ID, TIV2015 is the total investment value in 2015, TIV2016 is the total investment value in 2016, LAT is the latitude of the policy holder's city, and LON is the longitude of the policy holder's city.

Sample Input

TEXT

PID	TIV_2015	TIV_2016	LAT	LON
1	10	5	10	10
2	20	20	20	20
3	10	30	20	20
4	10	40	40	40

Sample Output

TEXT

TIV_2016
45.00

Explanation

TEXT

The first record in the table, like the last record, meets both of the two criteria. The TIV_2015 value '10' is the same as the third and fourth record, and its location is unique.

The second record does not meet any of the two criteria. Its TIV_2015 is not like the first or last record, and its location is not unique.

And its location is the same with the third record, which makes the third record also not unique.

So, the result is the sum of TIV_2016 of the first and last record, which is 45.

Solution

SQL



```
SELECT SUM(TIV_2016) AS TIV_2016
FROM insurance
WHERE CONCAT(LAT, ',', LON)
      IN (SELECT CONCAT(LAT, ',', LON)
          FROM insurance
          GROUP BY LAT, LON
          HAVING COUNT(1) = 1)
AND TIV_2015 IN
  (SELECT TIV_2015
   FROM insurance
   GROUP BY TIV_2015
   HAVING COUNT(1) > 1)
```

586. Customer Placing the Largest Number of Orders | Easy | [LeetCode](#)

Query the customer_number from the orders table for the customer who has placed the largest number of orders.

It is guaranteed that exactly one customer will have placed more orders than any other customer.

The orders table is defined as follows:

TEXT

Column	Type
order_number (PK)	int
customer_number	int
order_date	date
required_date	date
shipped_date	date
status	char(15)
comment	char(200)

Sample Input

TEXT

order_number	customer_number	order_date	required_date	shipped_date	st
1	1	2017-04-09	2017-04-13	2017-04-12	C1
2	2	2017-04-15	2017-04-20	2017-04-18	C1
3	3	2017-04-16	2017-04-25	2017-04-20	C1
4	3	2017-04-18	2017-04-28	2017-04-25	C1

Sample Output

TEXT

customer_number
3

Explanation

TEXT

The customer with number '3' has two orders, which is greater than either customer
So the result is customer_number '3'.

Solution

SQL



```
# assume: only one match
```

```
SELECT customer_number FROM orders  
GROUP BY customer_number  
ORDER BY COUNT(1) DESC  
LIMIT 1
```

```
## assume: multiple matches
```

```
## . 1 1
```

```
## . 2 1
```

```
## . 3 1
```

```
##
```

```
## . 1 1 1 1
```

```
## . 1 1 2 1
```

```
## . 1 1 3 1
```

```
##
```

```
## . SELECT t1.customer_number
```

```
## . FROM (SELECT customer_number, COUNT(1) AS count
```

```
## . FROM orders GROUP BY customer_number) AS t1,
```

```
## . (SELECT customer_number, COUNT(1) AS count
```

```
## . FROM orders GROUP BY customer_number) AS t2
```

```
## . GROUP BY t1.customer_number
```

```
## . HAVING max(t1.count) = max(t2.count)
```

595. Big Countries | Easy | [LeetCode](#)

There is a table world

TEXT

name	continent	area	population	gdp
Afghanistan	Asia	652230	25500100	20343000
Albania	Europe	28748	2831741	12960000
Algeria	Africa	2381741	37100000	188681000
Andorra	Europe	468	78115	3712000
Angola	Africa	1246700	20609294	100990000

A country is big if it has an area of bigger than 3 million square km or a population of more than 25 million.

Write a SQL solution to output big countries' name, population and area.

For example, according to the above table, we should output:

TEXT

name	population	area
Afghanistan	25500100	652230
Algeria	37100000	2381741

Solution

SQL

```
SELECT name, population, area
FROM World
WHERE area >= 3000000 OR population > 25000000;
```



There is a table `courses` with columns: **student** and **class**

Please list out all classes which have more than or equal to 5 students.

For example, the table:

TEXT

+-----+-----+		
student	class	
+-----+-----+		
A	Math	
B	English	
C	Math	
D	Biology	
E	Math	
F	Computer	
G	Math	
H	Math	
I	Math	
+-----+-----+		

Should output:

TEXT

+-----+		
class		
+-----+		
Math		
+-----+		

Solution

SQL

```
SELECT class
FROM courses
GROUP BY class
HAVING count(DISTINCT Student)>=5;
```



597. Friend Requests I: Overall Acceptance Rate | Easy |

[LeetCode](#)

In social network like Facebook or Twitter, people send friend requests and accept others' requests as well. Now given two tables as below: Table: friend_request

TEXT

sender_id	send_to_id	request_date
1	2	2016_06-01
1	3	2016_06-01
1	4	2016_06-01
2	3	2016_06-02
3	4	2016-06-09

Table: request_accepted

TEXT

requester_id	accepter_id	accept_date
1	2	2016_06-03
1	3	2016-06-08
2	3	2016-06-08
3	4	2016-06-09
3	4	2016-06-10

Write a query to find the overall acceptance rate of requests rounded to 2 decimals, which is the number of acceptance divide the number of requests. For the sample data above, your query should return the following result.

TEXT

accept_rate
0.80

Note:

The accepted requests are not necessarily from the table friendrequest. *In this case, you just need to simply count the total accepted requests (no matter whether they are in the original requests), and divide it by the number of requests to get the acceptance rate. It is possible that a sender sends multiple requests to the same receiver, and a request could be accepted more than once. In this case, the 'duplicated' requests or acceptances are only counted once. If there is no requests at all, you should return 0.00 as the acceptance rate.* Explanation: There are 4 unique accepted requests, and there are 5 requests in total. So the rate is 0.80.

Follow-up:

Can you write a query to return the accept rate but for every month? How about the cumulative accept rate for every day?

Solution

SQL



```
SELECT IFNULL((round(accepts/requests, 2)), 0.0) AS accept_rate
FROM
    (SELECT count(DISTINCT sender_id, send_to_id) AS requests FROM friend_request
    (SELECT count(DISTINCT requester_id, accepter_id) AS accepts FROM request_acc
```

601. Human Traffic of Stadium | Hard | [LeetCode](#)

Table: Stadium

TEXT

+	-----	+	-----	+
	Column Name		Type	
+	-----	+	-----	+
	id		int	
	visit_date		date	
	people		int	
+	-----	+	-----	+

visitdate is the primary key for this table. Each row of this table contains the visit date and visit id to the stadium with the number of people during the visit. No two rows will have the same visitdate, and as the id increases, the dates increase as well.

Write an SQL query to display the records with three or more rows with **consecutive** id's, and the number of people is greater than or equal to 100 for each.

Return the result table ordered by `visit_date` in **ascending order**.

The query result format is in the following example.

TEXT

Stadium table:

id	visit_date	people
1	2017-01-01	10
2	2017-01-02	109
3	2017-01-03	150
4	2017-01-04	99
5	2017-01-05	145
6	2017-01-06	1455
7	2017-01-07	199
8	2017-01-09	188

Result table:

id	visit_date	people
5	2017-01-05	145
6	2017-01-06	1455
7	2017-01-07	199
8	2017-01-09	188

The four rows with ids 5, 6, 7, and 8 have consecutive ids and each of them has > 100 people. The rows with ids 2 and 3 are not included because we need at least three consecutive

Solution

SQL



```
SELECT DISTINCT s1.*
FROM Stadium s1 JOIN Stadium s2 JOIN Stadium s3
ON (s1.id = s2.id-1 AND s1.id = s3.id-2) OR
(s1.id = s2.id+1 AND s1.id = s3.id-1) OR
(s1.id = s2.id+1 AND s1.id = s3.id+2)
WHERE s1.people >= 100 AND s2.people >= 100 AND s3.people >= 100
ORDER BY visit_date
```

602. Friend Requests II: Who Has the Most Friends | Medium | [LeetCode](#)

In social network like Facebook or Twitter, people send friend requests and accept others' requests as well. Table `request_accepted` holds the data of friend acceptance, while `requesterid` and `accepterid` both are the id of a person.

TEXT

requester_id	accepter_id	accept_date
1	2	2016-06-03
1	3	2016-06-08
2	3	2016-06-08
3	4	2016-06-09

Write a query to find the people who have the most friends and the most friends number. For the sample data above, the result is:

TEXT

id	num
3	3

Note:

It is guaranteed there is only 1 people having the most friends. The friend request could only been accepted once, which mean there is no multiple records with the same requesterid and accepterid value. Explanation: The person with id '3' is a friend of people '1', '2' and '4', so he has 3 friends in total, which is the most number than any others.

Follow-up: In the real world, multiple people could have the same most number of friends, can you find all these people in this case?

SQL



```
SELECT t.id, sum(t.num) AS num
FROM (
    (SELECT requester_id AS id, COUNT(1) AS num
     FROM request_accepted
     GROUP BY requester_id)
  union all
    (SELECT accepter_id AS id, COUNT(1) AS num
     FROM request_accepted
     GROUP BY accepter_id)) AS t
GROUP BY t.id
ORDER BY num DESC
LIMIT 1;
```

603. Consecutive Available Seats | Easy | [LeetCode](#)

Several friends at a cinema ticket office would like to reserve consecutive available seats. Can you help to query all the consecutive available seats order by the seat_id using the following cinema table?

TEXT

seat_id	free
1	1
2	0
3	1

4	1	
5	1	

Your query should return the following result for the sample case above.

TEXT

seat_id

3
4
5

Note:

The seat_id is an auto increment int, and free is bool ('1' means free, and '0' means occupied.). Consecutive available seats are more than 2(inclusive) seats consecutively available.

Solution

SQL

```
SELECT DISTINCT t1.seat_id
FROM cinema AS t1 JOIN cinema AS t2
ON abs(t1.seat_id-t2.seat_id)=1
WHERE t1.free='1' AND t2.free='1'
ORDER BY t1.seat_id
```



607.Sales Person | Easy | [LeetCode](#)

Description

Given three tables: salesperson, company, orders. Output all the names in the table salesperson, who didn't have sales to company 'RED'.

Example Input

Table: salesperson

TEXT

sales_id	name	salary	commission_rate	hire_date
1	John	100000	6	4/1/2006
2	Amy	120000	5	5/1/2010
3	Mark	65000	12	12/25/2008
4	Pam	25000	25	1/1/2005
5	Alex	50000	10	2/3/2007

The table salesperson holds the salesperson information. Every salesperson has a sales_id and a name. Table: company

TEXT

com_id	name	city
1	RED	Boston
2	ORANGE	New York
3	YELLOW	Boston
4	GREEN	Austin

The table company holds the company information. Every company has a com_id and a name. Table: orders

TEXT

order_id	date	com_id	sales_id	amount
1	1/1/2014	3	4	100000
2	2/1/2014	4	5	5000
3	3/1/2014	1	1	50000
4	4/1/2014	1	4	25000

The table orders holds the sales record information, salesperson and customer company are represented by *salesid* and *comid*. output

TEXT

```
+-----+
| name |
+-----+
| Amy  |
| Mark |
| Alex |
+-----+
```

Explanation

According to order '3' and '4' in table orders, it is easy to tell only salesperson 'John' and 'Alex' have sales to company 'RED', so we need to output all the other names in table salesperson.

Solution

SQL

```
SELECT name
FROM salesperson
WHERE name NOT IN
    (SELECT DISTINCT salesperson.name
     FROM salesperson, orders, company
     WHERE company.name = 'RED'
     AND salesperson.sales_id = orders.sales_id
     AND orders.com_id = company.com_id)
```



608. Tree Node | Medium | [LeetCode](#)

Given a table tree, id is identifier of the tree node and p_id is its parent node's id.

TEXT

+-----+-----+	
id	p_id
+-----+-----+	
1	null
2	1
3	1
4	2
5	2
+-----+-----+	

Each node in the tree can be one of three types:

Leaf: if the node is a leaf node. Root: if the node is the root of the tree. Inner: If the node is neither a leaf node nor a root node. Write a query to print the node id and the type of the node. Sort your output by the node id. The result for the above sample is:

TEXT

+-----+-----+	
id	Type
+-----+-----+	
1	Root
2	Inner
3	Leaf
4	Leaf
5	Leaf
+-----+-----+	

Explanation

Node '1' is root node, because its parent node is NULL and it has child node '2' and '3'. Node '2' is inner node, because it has parent node '1' and child node '4' and '5'. Node '3', '4' and '5' is Leaf node, because they have parent node and they don't have child node. And here is the image of the sample tree as below:

TEXT



Note

If there is only one node on the tree, you only need to output its root attributes.

Solution

SQL



```
## Basic Ideas: LEFT JOIN
# In tree, each node can only one parent or no parent
## | id | p_id | id (child) |
## |-----|
## |..1 | null |.....1 |
## |..1 | null |.....2 |
## |..2 |...1 |.....4 |
## |..2 |...1 |.....5 |
## |..3 |...1 |.....null |
## |..4 |...2 |.....null |
## |..5 |...2 |.....null |
```

```
SELECT t1.id,
       CASE
         WHEN ISNULL(t1.p_id) THEN 'Root'
         WHEN ISNULL(MAX(t2.id)) THEN 'Leaf'
         ELSE 'Inner'
       END AS Type
FROM tree AS t1 LEFT JOIN tree AS t2
ON t1.id = t2.p_id
GROUP BY t1.id, t1.p_id
```

610. Triangle Judgement | Easy | [LeetCode](#)

A pupil Tim gets homework to identify whether three line segments could possibly form a triangle. However, this assignment is very heavy because there are hundreds of records to calculate. Could you help Tim by writing a query to judge whether these three sides can form a triangle, assuming table triangle holds the length of the three sides x, y and z.

TEXT

x	y	z
13	15	30
10	20	15

For the sample data above, your query should return the follow result:

TEXT

x	y	z	triangle
13	15	30	No
10	20	15	Yes

Solution

SQL

```
SELECT x, y, z,
       CASE
         WHEN x+y>z AND y+z>x AND x+z>y THEN 'Yes'
         ELSE 'No'
       END AS triangle
FROM triangle
```



612. Shortest Distance in a Plane | Medium | [LeetCode](#)

Table point_2d holds the coordinates (x,y) of some unique points (more than two) in a plane. Write a query to find the shortest distance between these points rounded to 2 decimals.

TEXT

x	y
-1	-1
0	0
-1	-2

The shortest distance is 1.00 from point (-1,-1) to (-1,2). So the output should be:

TEXT

shortest
1.00

Note: The longest distance among all the points are less than 10000.

Solution

SQL

```
SELECT ROUND(MIN(SQRT((t1.x-t2.x)*(t1.x-t2.x) + (t1.y-t2.y)*(t1.y-t2.y))), 2) AS shortest
FROM point_2d AS t1, point_2d AS t2
WHERE t1.x!=t2.x OR t1.y!=t2.y
```

```
# SELECT ROUND(SQRT((t1.x-t2.x)*(t1.x-t2.x) + (t1.y-t2.y)*(t1.y-t2.y)), 2) AS shortest
# FROM point_2d AS t1, point_2d AS t2
# WHERE t1.x!=t2.x OR t1.y!=t2.y
# ORDER BY shortest ASC
# LIMIT 1
```

Table point holds the x coordinate of some points on x-axis in a plane, which are all integers. Write a query to find the shortest distance between two points in these points.

TEXT

	x	

	-1	
	0	
	2	

The shortest distance is '1' obviously, which is from point '-1' to '0'. So the output is as below:

TEXT

	shortest	

	1	

Note: Every point is unique, which means there is no duplicates in table point.

Follow-up: What if all these points have an id and are arranged from the left most to the right most of x axis?

Solution

SQL

```
SELECT t1.x-t2.x AS shortest
FROM point AS t1 JOIN point AS t2
WHERE t1.x>t2.x
ORDER BY (t1.x-t2.x) ASC
LIMIT 1
```



In facebook, there is a follow table with two columns: followee, follower.

Please write a sql query to get the amount of each follower's follower if he/she has one.

For example:

TEXT

+-----+-----+	
followee	follower
+-----+-----+	
A	B
B	C
B	D
D	E
+-----+-----+	

should output:

TEXT

+-----+-----+	
follower	num
+-----+-----+	
B	2
D	1
+-----+-----+	

Explanation: Both B and D exist in the follower list, when as a followee, B's follower is C and D, and D's follower is E. A does not exist in follower list.

Note: Followee would not follow himself/herself in all cases. Please display the result in follower's alphabet order.

Solution



SQL

```
## Explain the business logic
## ... A follows B. Then A is follwer, B is followee
## What are second degree followers?
## ... A follows B, and B follows C.
## ... Then A is the second degree followers of C

SELECT f1.follower, COUNT(DISTINCT f2.follower) AS num
FROM follow AS f1 JOIN follow AS f2
ON f1.follower = f2.followee
GROUP BY f1.follower;
```

615. Average Salary: Departments VS Company | Hard |

LeetCode

Given two tables as below, write a query to display the comparison result (higher/lower/same) of the average salary of employees in a department to the company's average salary. Table: salary

TEXT

id	employee_id	amount	pay_date
1	1	9000	2017-03-31
2	2	6000	2017-03-31
3	3	10000	2017-03-31
4	1	7000	2017-02-28
5	2	6000	2017-02-28
6	3	8000	2017-02-28

The *employeeid* column refers to the *employeeid* in the following table employee.

TEXT

employee_id	department_id
1	1
2	2
3	2

So for the sample data above, the result is:

TEXT

pay_month	department_id	comparison
2017-03	1	higher
2017-03	2	lower
2017-02	1	same
2017-02	2	same

Explanation In March, the company's average salary is $(9000+6000+10000)/3 = 8333.33...$ The average salary for department '1' is 9000, which is the salary of employeeid '1' since there is only one employee in this department. So the comparison result is 'higher' since $9000 > 8333.33$ obviously. The average salary of department '2' is $(6000 + 10000)/2 = 8000$, which is the average of employeeid '2' and '3'. So the comparison result is 'lower' since $8000 < 8333.33$. With the same formula for the average salary comparison in February, the result is 'same' since both the department '1' and '2' have the same average salary with the company, which is 7000.

Solution

SQL



```
SELECT t1.pay_month, t1.department_id,
       (CASE WHEN t1.amount = t2.amount THEN 'same'
            WHEN t1.amount > t2.amount THEN 'higher'
            WHEN t1.amount < t2.amount THEN 'lower' END) AS comparison
FROM
  (SELECT left(pay_date, 7) AS pay_month, department_id, avg(amount) AS amount
   FROM salary JOIN employee
   ON salary.employee_id = employee.employee_id
   GROUP BY pay_month, department_id
   ORDER BY pay_month DESC, department_id) AS t1
JOIN
  (SELECT left(pay_date, 7) AS pay_month, avg(amount) AS amount
   FROM salary JOIN employee
```

```
ON salary.employee_id = employee.employee_id
GROUP BY pay_month) AS t2
ON t1.pay_month = t2.pay_month
```

□

618. Students Report By Geography | Hard | [LeetCode](#)

A U.S graduate school has students from Asia, Europe and America. The students' location information are stored in table student as below.

TEXT

name	continent
Jack	America
Pascal	Europe
Xi	Asia
Jane	America

Pivot the continent column in this table so that each name is sorted alphabetically and displayed underneath its corresponding continent. The output headers should be America, Asia and Europe respectively. It is guaranteed that the student number from America is no less than either Asia or Europe. For the sample input, the output is:

TEXT

America	Asia	Europe
Jack	Xi	Pascal
Jane		

Follow-up: If it is unknown which continent has the most students, can you write a query to generate the student report?

Solution

```

SELECT t1.name AS America, t2.name AS Asia, t3.name AS Europe
FROM
    (SELECT (@cnt1 := @cnt1 + 1) AS id, name
    FROM student
    CROSS JOIN (SELECT @cnt1 := 0) AS dummy
    WHERE continent='America'
    ORDER BY name) AS t1
LEFT JOIN
    (SELECT (@cnt2 := @cnt2 + 1) AS id, name
    FROM student
    CROSS JOIN (SELECT @cnt2 := 0) AS dummy
    WHERE continent='Asia'
    ORDER BY name) AS t2
ON t1.id = t2.id
LEFT JOIN
    (SELECT (@cnt3 := @cnt3 + 1) AS id, name
    FROM student
    CROSS JOIN (SELECT @cnt3 := 0) AS dummy
    WHERE continent='Europe'
    ORDER BY name) AS t3
ON t1.id = t3.id

```

619. Biggest Single Number | Easy | [LeetCode](#)

Table number contains many numbers in column num including duplicated ones. Can you write a SQL query to find the biggest number, which only appears once.

TEXT

```

+----+
| num |
+----+
| 8 |
| 8 |
| 3 |
| 3 |
| 1 |
| 4 |

```

```
| 5 |  
| 6 |
```

For the sample data above, your query should return the following result:

```
TEXT  
+----+  
| num |  
+----+  
| 6   |
```

Note: If there is no such number, just output null.

Solution

```
SQL  
SELECT IFNULL(  
    SELECT num  
    FROM number  
    GROUP BY num  
    HAVING count(1) = 1  
    ORDER BY num DESC  
    LIMIT 0, 1), NULL) AS num
```



620. Not Boring Movies | Easy | [LeetCode](#)

X city opened a new cinema, many people would like to go to this cinema. The cinema also gives out a poster indicating the movies' ratings and descriptions. Please write a SQL query to output movies with an odd numbered ID and a description that is not 'boring'. Order the result by rating.

For example, table cinema:

TEXT

id	movie	description	rating
1	War	great 3D	8.9
2	Science	fiction	8.5
3	irish	boring	6.2
4	Ice song	Fantasy	8.6
5	House card	Interesting	9.1

For the example above, the output should be:

TEXT

id	movie	description	rating
5	House card	Interesting	9.1
1	War	great 3D	8.9

Solution

SQL

```
SELECT *
FROM Cinema
WHERE description <> 'boring' AND ID % 2 = 1
ORDER BY rating DESC;
```



626. Exchange Seats | Medium | [LeetCode](#)

Mary is a teacher in a middle school and she has a table `seat` storing students' names and their corresponding seat ids.

The column `id` is continuous increment.

Mary wants to change seats for the adjacent students.

Can you write a SQL query to output the result for Mary?

TEXT

+-----+-----+	
id student	
+-----+-----+	
1 Abbot	
2 Doris	
3 Emerson	
4 Green	
5 Jeames	
+-----+-----+	

For the sample input, the output is:

TEXT

+-----+-----+	
id student	
+-----+-----+	
1 Doris	
2 Abbot	
3 Green	
4 Emerson	
5 Jeames	
+-----+-----+	

Note:

If the number of students is odd, there is no need to change the last one's seat.

Solution

SQL

SELECT

IF(id<(SELECT MAX(id) FROM seat),IF(id%2=0,id-1, id+1),IF(id%2=0, id-1, id)) AS id




```
FROM seat
ORDER BY id;
```

627. Swap Salary | [LeetCode](#)

Table: Salary

TEXT

+-----+-----+		
Column Name	Type	
+-----+-----+		
id	int	
name	varchar	
sex	ENUM	
salary	int	
+-----+-----+		

id is the primary key for this table.

The sex column is ENUM value of type ('m', 'f').

The table contains information about an employee.

Write an SQL query to swap all 'f' and 'm' values (i.e., change all 'f' values to 'm' and vice versa) with a single update statement and no intermediate temp table(s).

Note that you must write a single update statement, DO NOT write any select statement for this problem.

The query result format is in the following example:

TEXT

Salary table:

+----+-----+-----+-----+				
id	name	sex	salary	
+----+-----+-----+-----+				
1	A	m	2500	

	2		B		f		1500	
	3		C		m		5500	
	4		D		f		500	

+-----+-----+-----+-----+

Result table:

	id		name		sex		salary	
--	----	--	------	--	-----	--	--------	--

+-----+-----+-----+-----+

	1		A		f		2500	
	2		B		m		1500	
	3		C		f		5500	
	4		D		m		500	

+-----+-----+-----+-----+

(1, A) and (2, C) were changed from 'm' to 'f'.
 (2, B) and (4, D) were changed from 'f' to 'm'.

Solution

SQL

With IF

```
UPDATE Salary SET sex = IF(sex='m', 'f', 'm')
```

With CASE

```
UPDATE Salary SET sex = CASE WHEN sex='m' THEN 'f' ELSE 'm' END
```

1045. Customers Who Bought All Products | Medium |

[LeetCode](#)

Table: Customer

TEXT

	Column Name		Type	
--	-------------	--	------	--

+-----+-----+

	customer_id		int	
--	-------------	--	-----	--

```

| product_key | int      |
+-----+

```

product_key is a foreign key to Product table. Table: Product

TEXT

```

+-----+
| Column Name | Type      |
+-----+
| product_key | int      |
+-----+

```

product_key is the primary key column for this table.

Write an SQL query for a report that provides the customer ids from the Customer table that bought all the products in the Product table.

For example:

TEXT

Customer table:

```

+-----+
| customer_id | product_key |
+-----+
| 1           | 5           |
| 2           | 6           |
| 3           | 5           |
| 3           | 6           |
| 1           | 6           |
+-----+

```

Product table:

```

+-----+
| product_key |
+-----+
| 5           |
| 6           |
+-----+

```

Result table:

```
+-----+
| customer_id |
+-----+
| 1           |
| 3           |
+-----+
```

The customers who bought all the products (5 and 6) are customers with id 1 and 3

Solution

SQL

```
SELECT customer_id
FROM Customer
GROUP BY customer_id
HAVING count(DISTINCT product_key) = (
    SELECT count(1)
    FROM Product)
```



1050. Actors and Directors Who Cooperated At Least Three Times | Easy | [LeetCode](#)

Table: ActorDirector

TEXT

```
+-----+-----+
| Column Name | Type   |
+-----+-----+
| actor_id    | int    |
| director_id | int    |
| timestamp   | int    |
+-----+-----+
```

timestamp is the primary key column for this table.

Write a SQL query for a report that provides the pairs (*actorid*, *directorid*) where the actor have cooperated with the director at least 3 times.

Example:

TEXT

ActorDirector table:

actor_id	director_id	timestamp
1	1	0
1	1	1
1	1	2
1	2	3
1	2	4
2	1	5
2	1	6

Result table:

actor_id	director_id
1	1

The only pair is (1, 1) where they cooperated exactly 3 times.

Solution

SQL

```
SELECT actor_id, director_id
FROM ActorDirector
GROUP BY actor_id, director_id
HAVING COUNT(1) >= 3
```



Table: Sales

TEXT

+-----+-----+		
Column Name	Type	
+-----+-----+		
sale_id	int	
product_id	int	
year	int	
quantity	int	
price	int	
+-----+-----+		

(sale_id, year) is the primary key of this table.

product_id is a foreign key to Product table.

Note that the price is per unit.

Table: Product

TEXT

+-----+-----+		
Column Name	Type	
+-----+-----+		
product_id	int	
product_name	varchar	
+-----+-----+		

product_id is the primary key of this table.

Write an SQL query that reports all product names of the products in the Sales table along with their selling year and price.

For example:

TEXT

Sales table:

+-----+-----+-----+-----+				
sale_id	product_id	year	quantity	price
+-----+-----+-----+-----+				
1	100	2008	10	5000

2	100	2009	12	5000
7	200	2011	15	9000

Product table:

product_id	product_name
100	Nokia
200	Apple
300	Samsung

Result table:

product_name	year	price
Nokia	2008	5000
Nokia	2009	5000
Apple	2011	9000

Solution

```

SQL
SELECT product_name, year, price
FROM Sales JOIN Product
ON Product.product_id = Sales.product_id

```



1069. Product Sales Analysis II | Easy | [LeetCode](#)

Table: Sales

TEXT

Column Name	Type
-------------	------

```

+-----+-----+
| sale_id | int |
| product_id | int |
| year | int |
| quantity | int |
| price | int |
+-----+-----+

```

sale_id is the primary key of this table.
product_id is a foreign key to Product table.
Note that the price is per unit.

Table: Product

TEXT

```

+-----+-----+
| Column Name | Type |
+-----+-----+
| product_id | int |
| product_name | varchar |
+-----+-----+

```

product_id is the primary key of this table.

Write an SQL query that reports the total quantity sold for every product id.

The query result format is in the following example:

TEXT

Sales table:

```

+-----+-----+-----+-----+-----+
| sale_id | product_id | year | quantity | price |
+-----+-----+-----+-----+-----+
| 1 | 100 | 2008 | 10 | 5000 |
| 2 | 100 | 2009 | 12 | 5000 |
| 7 | 200 | 2011 | 15 | 9000 |
+-----+-----+-----+-----+-----+

```

Product table:

```

+-----+-----+

```


product_id	product_name
100	Nokia
200	Apple
300	Samsung

Result table:

product_id	total_quantity
100	22
200	15

Solution

SQL

```
SELECT product_id, sum(quantity) AS total_quantity
FROM Sales
GROUP BY product_id;
```



1070. Product Sales Analysis III | Medium | [LeetCode](https://leetcode.com/problems/product-sales-analysis-iii/)

Table: Sales

TEXT

Column Name	Type
sale_id	int
product_id	int
year	int
quantity	int
price	int

sale_id is the primary key of this table.
product_id is a foreign key to Product table.
Note that the price is per unit.

Table: Product

TEXT

Column Name	Type
product_id	int
product_name	varchar

product_id is the primary key of this table.

Write an SQL query that selects the product id, year, quantity, and price for the first year of every product sold.

The query result format is in the following example:

TEXT

Sales table:

sale_id	product_id	year	quantity	price
1	100	2008	10	5000
2	100	2009	12	5000
7	200	2011	15	9000

Product table:

product_id	product_name
100	Nokia
200	Apple
300	Samsung

Result table:

product_id	first_year	quantity	price
100	2008	10	5000
200	2011	15	9000

Solution

SQL

```
SELECT
    product_id,
    year first_year,
    quantity,
    price
FROM Sales
WHERE (product_id, year) IN (SELECT product_id, MIN(year)
                             FROM Sales
                             GROUP BY product_id)
```



1075. Project Employees I | Easy | [LeetCode](#)

Table: Project

TEXT

Column Name	Type
project_id	int
employee_id	int

(project_id, employee_id) is the primary key of this table.
employee_id is a foreign key to Employee table.

Table: Employee

TEXT

Column Name	Type
employee_id	int
name	varchar
experience_years	int

employee_id is the primary key of this table.

Write an SQL query that reports the average experience years of all the employees for each project, rounded to 2 digits.

The query result format is in the following example:

TEXT

Project table:

project_id	employee_id
1	1
1	2
1	3
2	1
2	4

Employee table:

employee_id	name	experience_years
1	Khaled	3
2	Ali	2
3	John	1
4	Doe	2

Result table:

+-----+-----+		
project_id	average_years	
+-----+-----+		
1	2.00	
2	2.50	
+-----+-----+		

The average experience years for the first project is $(3 + 2 + 1) / 3 = 2.00$ and

Solution

SQL



```
SELECT
    p.project_id,
    ROUND(AVG(e.experience_years),2) average_years
FROM
    Project p JOIN Employee e ON
    p.employee_id = e.employee_id
GROUP BY
    p.project_id
```

1076. Project Employees II | Easy | [LeetCode](#)

Table: Project

TEXT

+-----+-----+		
Column Name	Type	
+-----+-----+		
project_id	int	
employee_id	int	
+-----+-----+		

(project_id, employee_id) is the primary key of this table.

employee_id is a foreign key to Employee table.

Table: Employee

TEXT

Column Name	Type
employee_id	int
name	varchar
experience_years	int

employee_id is the primary key of this table.

Write an SQL query that reports all the projects that have the most employees.

The query result format is in the following example:

TEXT

Project table:

project_id	employee_id
1	1
1	2
1	3
2	1
2	4

Employee table:

employee_id	name	experience_years
1	Khaled	3
2	Ali	2
3	John	1
4	Doe	2

Result table:

```

+-----+
| project_id |
+-----+
| 1          |
+-----+

```

The first project has 3 employees while the second one has 2.

SQL

```

SELECT project_id
FROM Project
GROUP BY project_id
HAVING COUNT(employee_id) = (SELECT COUNT(employee_id)
                             FROM Project
                             GROUP BY project_id
                             ORDER BY COUNT(employee_id) DESC
                             LIMIT 1)

```

1077. Project Employees III | Medium | [LeetCode](#)

Table: Project

TEXT

```

+-----+-----+
| Column Name | Type   |
+-----+-----+
| project_id  | int    |
| employee_id | int    |
+-----+-----+

```

(project_id, employee_id) is the primary key of this table.
employee_id is a foreign key to Employee table.

Table: Employee

TEXT

Column Name	Type
employee_id	int
name	varchar
experience_years	int

employee_id is the primary key of this table.

Write an SQL query that reports the most experienced employees in each project. In case of a tie, report all employees with the maximum number of experience years.

The query result format is in the following example:

TEXT

Project table:

project_id	employee_id
1	1
1	2
1	3
2	1
2	4

Employee table:

employee_id	name	experience_years
1	Khaled	3
2	Ali	2
3	John	3
4	Doe	2

Result table:

project_id	employee_id
------------	-------------

project_id	employee_id
1	1
1	3
2	1

Both employees with id 1 and 3 have the most experience among the employees of th

Solution

SQL



```
SELECT
    p.project_id,
    e.employee_id
FROM
    Project p LEFT JOIN Employee e ON
        p.employee_id = e.employee_id
WHERE (p.project_id,
        e.experience_years) IN (SELECT
                                p.project_id,
                                MAX(e.experience_years)
                                FROM
                                    Project p JOIN Employee e ON
                                        p.employee_id = e.employee_id
                                GROUP BY
                                    p.project_id)
```

1082. Sales Analysis I | Easy | [LeetCode](#)

Table: Product

TEXT

Column Name	Type
-------------	------

```

| product_id | int |
| product_name | varchar |
| unit_price | int |
+-----+

```

product_id is the primary key of this table.

Table: Sales

TEXT

```

+-----+
| Column Name | Type |
+-----+
| seller_id | int |
| product_id | int |
| buyer_id | int |
| sale_date | date |
| quantity | int |
| price | int |
+-----+

```

This table has no primary key, it can have repeated rows.
product_id is a foreign key to Product table.

Write an SQL query that reports the best seller by total sales price, If there is a tie, report them all.

The query result format is in the following example:

TEXT

Product table:

```

+-----+
| product_id | product_name | unit_price |
+-----+
| 1 | S8 | 1000 |
| 2 | G4 | 800 |
| 3 | iPhone | 1400 |
+-----+

```

Sales table:

seller_id	product_id	buyer_id	sale_date	quantity	price
1	1	1	2019-01-21	2	2000
1	2	2	2019-02-17	1	800
2	2	3	2019-06-02	1	800
3	3	4	2019-05-13	2	2800

Result table:

seller_id
1
3

Both sellers with id 1 and 3 sold products with the most total price of 2800.

Solution

SQL

```
SELECT seller_id
FROM Sales
GROUP BY seller_id
HAVING SUM(price) = (SELECT SUM(price)
                     FROM Sales
                     GROUP BY seller_id
                     ORDER BY SUM(price) DESC
                     LIMIT 1)
```



1083. Sales Analysis II | Easy | [LeetCode](#)

Table: Product

TEXT

+-----+-----+		
Column Name	Type	
+-----+-----+		
product_id	int	
product_name	varchar	
unit_price	int	
+-----+-----+		

product_id is the primary key of this table.

Table: Sales

TEXT

+-----+-----+		
Column Name	Type	
+-----+-----+		
seller_id	int	
product_id	int	
buyer_id	int	
sale_date	date	
quantity	int	
price	int	
+-----+-----+		

This table has no primary key, it can have repeated rows.

product_id is a foreign key to Product table.

Write an SQL query that reports the buyers who have bought S8 but not iPhone.

Note that S8 and iPhone are products present in the Product table.

The query result format is in the following example:

TEXT

Product table:

+-----+-----+-----+		
product_id	product_name	unit_price
+-----+-----+-----+		
1	S8	1000
2	G4	800

3	iPhone	1400
---	--------	------

Sales table:

seller_id	product_id	buyer_id	sale_date	quantity	price
1	1	1	2019-01-21	2	2000
1	2	2	2019-02-17	1	800
2	1	3	2019-06-02	1	800
3	3	3	2019-05-13	2	2800

Result table:

buyer_id
1

The buyer with id 1 bought an S8 but didn't buy an iPhone. The buyer with id 3 bo

Solution

SQL

```
SELECT DISTINCT s.buyer_id
FROM Sales s LEFT JOIN Product p ON
    s.product_id = p.product_id
WHERE p.product_name = 'S8' AND
    s.buyer_id NOT IN (SELECT s.buyer_id
                      FROM Sales s LEFT JOIN Product p ON
                          s.product_id = p.product_id
                      WHERE p.product_name = 'iPhone')
```



Reports the products that were only sold in spring 2019. That is, between 2019-01-01 and 2019-03-31 inclusive. Select the product that were only sold in spring 2019.

TEXT

Product table:

product_id	product_name	unit_price
1	S8	1000
2	G4	800
3	iPhone	1400

Sales table:

seller_id	product_id	buyer_id	sale_date	quantity	price
1	1	1	2019-01-21	2	2000
1	2	2	2019-02-17	1	800
2	2	3	2019-06-02	1	800
3	3	4	2019-05-13	2	2800

Result table:

product_id	product_name
1	S8

The product with id 1 was only sold in spring 2019 while the other two were sold

Solution

SQL

```
(SELECT DISTINCT s.product_id, p.product_name
FROM Sales s LEFT JOIN Product p ON
    s.product_id = p.product_id
```



```

WHERE s.sale_date >= '2019-01-01' AND
      s.sale_date <= '2019-03-31')
EXCEPT -- MINUS if Oracle
(SELECT DISTINCT s.product_id, p.product_name
FROM Sales s LEFT JOIN Product p ON
      s.product_id = p.product_id
WHERE s.sale_date < '2019-01-01' OR
      s.sale_date > '2019-03-31')

```

1097. Game Play Analysis V | Hard | [LeetCode](#)

We define the install date of a player to be the first login day of that player. We also define day 1 retention of some date X to be the number of players whose install date is X and they logged back in on the day right after X , divided by the number of players whose install date is X, rounded to 2 decimal places. Write an SQL query that reports for each install date, the number of players that installed the game on that day and the day 1 retention. The query result format is in the following example:

TEXT

Activity table:

player_id	device_id	event_date	games_played
1	2	2016-03-01	5
1	2	2016-03-02	6
2	3	2017-06-25	1
3	1	2016-03-01	0
3	4	2016-07-03	5

Result table:

install_dt	installs	Day1_retention
2016-03-01	2	0.50
2017-06-25	1	0.00

Player 1 and 3 installed the game on 2016-03-01 but only player 1 logged back in
Player 2 installed the game on 2017 -06-25 but didn't log back in on 2017-06-26 s

Solution

SQL



```
SELECT
    install_dt,
    COUNT(player_id) installs,
    ROUND(COUNT(retention)/COUNT(player_id),2) Day1_retention  --the number of re
FROM
    (
    SELECT a.player_id, a.install_dt, b.event_date retention -- id, the record of
    FROM
        (SELECT player_id, MIN(event_date) install_dt  --subquery 1 take the fir
        FROM Activity
        GROUP BY player_id) a LEFT JOIN Activity b ON  --sql left join the origi
            a.player_id = b.player_id AND
            a.install_dt + 1=b.event_date
    ) AS tmp
GROUP BY
    install_dt
```

1098. Unpopular Books | Medium | [LeetCode](#)

Table: Books

TEXT

```
+-----+-----+
| Column Name | Type |
+-----+-----+
| book_id     | int  |
| name       | varchar |
| available_from | date |
```



```
+-----+
book_id is the primary key of this table.
```

Table: orders

TEXT

```
+-----+
| Column Name | Type |
+-----+
| order_id    | int  |
| book_id     | int  |
| quantity    | int  |
| dispatch_date | date |
+-----+
order_id is the primary key of this table.
book_id is a foreign key to the Books table.
```

Write an SQL query that reports the books that have sold less than 10 copies in the last year, excluding books that have been available for less than 1 month from today. Assume today is 2019-06-23.

The query result format is in the following example:

TEXT

Books table:

```
+-----+-----+-----+
| book_id | name                | available_from |
+-----+-----+-----+
| 1       | "Kalila And Demna" | 2010-01-01     |
| 2       | "28 Letters"       | 2012-05-12     |
| 3       | "The Hobbit"       | 2019-06-10     |
| 4       | "13 Reasons Why"  | 2019-06-01     |
| 5       | "The Hunger Games" | 2008-09-21     |
+-----+-----+-----+
```

Orders table:

```
+-----+-----+-----+-----+
| order_id | book_id | quantity | dispatch_date |
```

book_id	name	quantity	dispatch_date
1	Kalila And Demna	1	2018-07-26
2	28 Letters	1	2018-11-05
3	The Hunger Games	8	2019-06-11
4	The Hunger Games	6	2019-06-05
5	The Hunger Games	5	2019-06-20
6	The Hunger Games	9	2009-02-02
7	The Hunger Games	8	2010-04-13

Result table:

book_id	name
1	"Kalila And Demna"
2	"28 Letters"
5	"The Hunger Games"

Solution

SQL



```

SELECT
    b.book_id, b.name
FROM
    Books b LEFT JOIN (
        -- subquery calculates last year's sales
        SELECT book_id, SUM(quantity) nsold
        FROM Orders
        WHERE dispatch_date BETWEEN '2018-06-23' AND '2019-06-23'
        GROUP BY book_id
    ) o
    ON b.book_id = o.book_id
WHERE
    (o.nsold < 10 OR o.nsold IS NULL) AND
    DATEDIFF('2019-06-23', b.available_from) > 30
    -- Sales less than 10 or no sales
    -- Not a new book within 1 month

```

1107. New Users Daily Count | Medium | [LeetCode](#)

Table: Traffic

TEXT

Column Name	Type
user_id	int
activity	enum
activity_date	date

There is no primary key for this table, it may have duplicate rows.

The activity column is an ENUM type of ('login', 'logout', 'jobs', 'groups', 'hom

Write an SQL query that reports for every date within at most 90 days from today, the number of users that logged in for the first time on that date. Assume today is 2019-06-30.

The query result format is in the following example:

TEXT

Traffic table:

user_id	activity	activity_date
1	login	2019-05-01
1	homepage	2019-05-01
1	logout	2019-05-01
2	login	2019-06-21
2	logout	2019-06-21
3	login	2019-01-01
3	jobs	2019-01-01
3	logout	2019-01-01
4	login	2019-06-21
4	groups	2019-06-21
4	logout	2019-06-21
5	login	2019-03-01

5	logout	2019-03-01
5	login	2019-06-21
5	logout	2019-06-21

Result table:

login_date	user_count
2019-05-01	1
2019-06-21	2

Note that we only care about dates with non zero user count.

The user with id 5 first logged in on 2019-03-01 so he's not counted on 2019-06-21



Solution

SQL



#Solution- 1:

```
SELECT login_date, COUNT(user_id) AS user_count
FROM (SELECT user_id, MIN(activity_date) AS login_date
      FROM Traffic
      WHERE activity = 'login'
      GROUP BY user_id) AS t
WHERE login_date >= DATE_ADD('2019-06-30', INTERVAL -90 DAY) AND login_date <= '2019-06-30'
GROUP BY login_date
```

#Solution- 2:

```
SELECT login_date, COUNT(user_id) user_count
FROM
  (SELECT user_id, MIN(activity_date) as login_date
   FROM Traffic
   WHERE activity='login'
   GROUP BY user_id) as t
WHERE DATEDIFF('2019-06-30', login_date) <= 90
GROUP BY login_date
```



1112. Highest Grade For Each Student | Medium | [LeetCode](https://leetcode.com/problems/highest-grade-for-each-student/)

Table: Enrollments

TEXT

Column Name	Type
student_id	int
course_id	int
grade	int

(student_id, course_id) is the primary key of this table.

Write a SQL query to find the highest grade with its corresponding course for each student. In case of a tie, you should find the course with the smallest courseid. *The output must be sorted by increasing studentid.*

The query result format is in the following example:

TEXT

Enrollments table:

student_id	course_id	grade
2	2	95
2	3	95
1	1	90
1	2	99
3	1	80
3	2	75
3	3	82

Result table:

student_id	course_id	grade
1	2	99
2	2	95

3	3	82
---	---	----

Solution

SQL

```
SELECT student_id, MIN(course_id) course_id, grade
FROM Enrollments
WHERE (student_id, grade) IN
      (SELECT student_id, MAX(grade)
       FROM Enrollments
       GROUP BY student_id)
GROUP BY student_id
ORDER BY student_id;
```

1113.Reported Posts | Easy | [LeetCode](#)

Table: Actions

TEXT

+-----+		
Column Name	Type	
+-----+		
user_id	int	
post_id	int	
action_date	date	
action	enum	
extra	varchar	
+-----+		

There is no primary key for this table, it may have duplicate rows.

The action column is an ENUM type of ('view', 'like', 'reaction', 'comment', 'rep

The extra column has optional information about the action such as a reason for r

Write an SQL query that reports the number of posts reported yesterday for each report reason. Assume today is 2019-07-05.

The query result format is in the following example:

TEXT

Actions table:

user_id	post_id	action_date	action	extra
1	1	2019-07-01	view	null
1	1	2019-07-01	like	null
1	1	2019-07-01	share	null
2	4	2019-07-04	view	null
2	4	2019-07-04	report	spam
3	4	2019-07-04	view	null
3	4	2019-07-04	report	spam
4	3	2019-07-02	view	null
4	3	2019-07-02	report	spam
5	2	2019-07-04	view	null
5	2	2019-07-04	report	racism
5	5	2019-07-04	view	null
5	5	2019-07-04	report	racism

Result table:

report_reason	report_count
spam	1
racism	2

Note that we only care about report reasons with non zero number of reports.

Solution



SQL

```
SELECT extra report_reason, COUNT(DISTINCT post_id) report_count
FROM
    (SELECT post_id, extra
    FROM Actions
    WHERE action_date = DATE_SUB('2019-07-05', INTERVAL 1 DAY) AND
        action = 'report') AS tmp
GROUP BY extra
```

1126. Active Businesses | Medium | [LeetCode](#)

Table: Events

TEXT

```
+-----+-----+
| Column Name | Type   |
+-----+-----+
| business_id | int    |
| event_type  | varchar|
| occurrences | int    |
+-----+-----+
```

(business_id, event_type) is the primary key of this table.

Each row in the table logs the info that an event of some type occurred at some bu

Write an SQL query to find all active businesses.

An active business is a business that has more than one event type with occurrences greater than the average occurrences of that event type among all businesses.

The query result format is in the following example:

TEXT

Events table:

```
+-----+-----+-----+
| business_id | event_type | occurrences |
+-----+-----+-----+
```


1	reviews	7	
3	reviews	3	
1	ads	11	
2	ads	7	
3	ads	6	
1	page views	3	
2	page views	12	

+-----+-----+-----+

Result table:

+-----+
business_id
+-----+
1
+-----+

Average for 'reviews', 'ads' and 'page views' are $(7+3)/2=5$, $(11+7+6)/3=8$, $(3+12)$
 Business with id 1 has 7 'reviews' events (more than 5) and 11 'ads' events (more

Solution

SQL



```
SELECT business_id
FROM (SELECT a.business_id, a.event_type, a.occurences, b.event_avg  -- sub 2
      FROM Events a LEFT JOIN
            (SELECT event_type, AVG(occurences) event_avg  -- sub 1
             FROM Events
             GROUP BY event_type) b ON
            a.event_type = b.event_type) tmp
WHERE occurences > event_avg
GROUP BY business_id
HAVING COUNT(event_type) > 1
```

1127. User Purchase Platform | Hard | [LeetCode](#)

Table: Spending

TEXT

+-----+-----+		
Column Name	Type	
+-----+-----+		
user_id	int	
spend_date	date	
platform	enum	
amount	int	
+-----+-----+		

The table logs the spendings history of users that make purchases from an online (user_id, spend_date, platform) is the primary key of this table.

The platform column is an ENUM type of ('desktop', 'mobile').

Write an SQL query to find the total number of users and the total amount spent using mobile only, desktop only and both mobile and desktop together for each date.

The query result format is in the following example:

TEXT

Spending table:

+-----+-----+-----+-----+				
user_id	spend_date	platform	amount	
+-----+-----+-----+-----+				
1	2019-07-01	mobile	100	
1	2019-07-01	desktop	100	
2	2019-07-01	mobile	100	
2	2019-07-02	mobile	100	
3	2019-07-01	desktop	100	
3	2019-07-02	desktop	100	
+-----+-----+-----+-----+				

Result table:

+-----+-----+-----+-----+				
spend_date	platform	total_amount	total_users	
+-----+-----+-----+-----+				
2019-07-01	desktop	100	1	
2019-07-01	mobile	100	1	

2019-07-01	both	200	1	
2019-07-02	desktop	100	1	
2019-07-02	mobile	100	1	
2019-07-02	both	0	0	

+-----+-----+-----+-----+

On 2019-07-01, user 1 purchased using both desktop and mobile, user 2 purchased u

On 2019-07-02, user 2 purchased using mobile only, user 3 purchased using desktop

Solution

SQL



```

SELECT aa.spend_date,
       aa.platform,
       COALESCE(bb.total_amount, 0) total_amount,
       COALESCE(bb.total_users,0) total_users
FROM
  (SELECT DISTINCT(spend_date), a.platform  -- table aa
   FROM Spending JOIN
     (SELECT 'desktop' AS platform UNION
      SELECT 'mobile' AS platform UNION
      SELECT 'both' AS platform
     ) a
   ) aa
LEFT JOIN
  (SELECT spend_date,                    -- table bb
       platform,
       SUM(amount) total_amount,
       COUNT(user_id) total_users
   FROM
     (SELECT spend_date,
              user_id,
              (CASE COUNT(DISTINCT platform)
                 WHEN 1 THEN platform
                 WHEN 2 THEN 'both'
                 END) platform,
              SUM(amount) amount
      FROM Spending

```

```

        GROUP BY spend_date, user_id
    ) b
GROUP BY spend_date, platform
) bb
ON aa.platform = bb.platform AND
aa.spend_date = bb.spend_date

```

1132. Reported Posts II | Medium | [LeetCode](#)

Table: Actions

TEXT

Column Name	Type
user_id	int
post_id	int
action_date	date
action	enum
extra	varchar

There is no primary key for this table, it may have duplicate rows.

The action column is an ENUM type of ('view', 'like', 'reaction', 'comment', 'rep

The extra column has optional information about the action such as a reason for r

Table: Removals

TEXT

Column Name	Type
post_id	int
remove_date	date

post_id is the primary key of this table.

Each row in this table indicates that some post was removed as a result of being

Write an SQL query to find the average for daily percentage of posts that got removed after being reported as spam, rounded to 2 decimal places.

The query result format is in the following example:

```
TEXT
Actions table:
+-----+-----+-----+-----+-----+
| user_id | post_id | action_date | action | extra |
+-----+-----+-----+-----+-----+
| 1      | 1      | 2019-07-01 | view  | null  |
| 1      | 1      | 2019-07-01 | like  | null  |
| 1      | 1      | 2019-07-01 | share | null  |
| 2      | 2      | 2019-07-04 | view  | null  |
| 2      | 2      | 2019-07-04 | report | spam  |
| 3      | 4      | 2019-07-04 | view  | null  |
| 3      | 4      | 2019-07-04 | report | spam  |
| 4      | 3      | 2019-07-02 | view  | null  |
| 4      | 3      | 2019-07-02 | report | spam  |
| 5      | 2      | 2019-07-03 | view  | null  |
| 5      | 2      | 2019-07-03 | report | racism |
| 5      | 5      | 2019-07-03 | view  | null  |
| 5      | 5      | 2019-07-03 | report | racism |
+-----+-----+-----+-----+-----+

Removals table:
+-----+-----+
| post_id | remove_date |
+-----+-----+
| 2      | 2019-07-20  |
| 3      | 2019-07-18  |
+-----+-----+

Result table:
+-----+
| average_daily_percent |
+-----+
| 75.00                  |
+-----+
```

+-----+

The percentage for 2019-07-04 is 50% because only one post of two spam reported p
The percentage for 2019-07-02 is 100% because one post was reported as spam and i
The other days had no spam reports so the average is $(50 + 100) / 2 = 75\%$
Note that the output is only one number and that we do not care about the remove

=====

Solution

SQL



```
WITH t1 AS(
SELECT a.action_date, (COUNT(DISTINCT r.post_id))/(COUNT(DISTINCT a.post_id)) AS
FROM (SELECT action_date, post_id
FROM actions
WHERE extra = 'spam' AND action = 'report') a
LEFT JOIN
removals r
ON a.post_id = r.post_id
GROUP BY a.action_date)

SELECT ROUND(AVG(t1.result)*100,2) AS average_daily_percent
FROM t1
```

=====

1141. User Activity for the Past 30 Days I | Easy | [LeetCode](#)

Table: Activity

TEXT

```
+-----+-----+
| Column Name | Type   |
+-----+-----+
| user_id     | int    |
| session_id  | int    |
| activity_date | date   |
| activity_type | enum   |
+-----+-----+
```

There is no primary key for this table, it may have duplicate rows.

The activity_type column is an ENUM of type ('open_session', 'end_session', 'scro

The table shows the user activities for a social media website.

Note that each session belongs to exactly one user.

Write an SQL query to find the daily active user count for a period of 30 days ending 2019-07-27 inclusively. A user was active on some day if he/she made at least one activity on that day.

The query result format is in the following example:

TEXT

Activity table:

user_id	session_id	activity_date	activity_type
1	1	2019-07-20	open_session
1	1	2019-07-20	scroll_down
1	1	2019-07-20	end_session
2	4	2019-07-20	open_session
2	4	2019-07-21	send_message
2	4	2019-07-21	end_session
3	2	2019-07-21	open_session
3	2	2019-07-21	send_message
3	2	2019-07-21	end_session
4	3	2019-06-25	open_session
4	3	2019-06-25	end_session

Result table:

day	active_users
2019-07-20	2
2019-07-21	2

Note that we do not care about days with zero active users.

Solution

SQL



```
SELECT activity_date AS day, COUNT(DISTINCT user_id) AS active_users
FROM activity
WHERE activity_date > '2019-06-26' AND activity_date < '2019-07-27'
GROUP BY activity_date
```

1142. User Activity for the Past 30 Days II | Easy | [LeetCode](#)

Table: Activity

TEXT

```
+-----+-----+
| Column Name | Type |
+-----+-----+
| user_id     | int  |
| session_id  | int  |
| activity_date | date |
| activity_type | enum |
+-----+-----+
```

There is no primary key for this table, it may have duplicate rows.

The activity_type column is an ENUM of type ('open_session', 'end_session', 'scro

The table shows the user activities for a social media website.

Note that each session belongs to exactly one user.

Write an SQL query to find the average number of sessions per user for a period of 30 days ending 2019-07-27 inclusively, rounded to 2 decimal places. The sessions we want to count for a user are those with at least one activity in that time period.

The query result format is in the following example:

TEXT

Activity table:

```
+-----+-----+-----+-----+
| user_id | session_id | activity_date | activity_type |
```


1	1	2019-07-20	open_session
1	1	2019-07-20	scroll_down
1	1	2019-07-20	end_session
2	4	2019-07-20	open_session
2	4	2019-07-21	send_message
2	4	2019-07-21	end_session
3	2	2019-07-21	open_session
3	2	2019-07-21	send_message
3	2	2019-07-21	end_session
3	5	2019-07-21	open_session
3	5	2019-07-21	scroll_down
3	5	2019-07-21	end_session
4	3	2019-06-25	open_session
4	3	2019-06-25	end_session

Result table:

average_sessions_per_user
1.33

User 1 and 2 each had 1 session in the past 30 days while user 3 had 2 sessions s

Solution

SQL

```
SELECT IFNULL(ROUND(AVG(a.num),2),0) AS average_sessions_per_user
FROM (
SELECT COUNT(DISTINCT session_id) AS num
FROM activity
WHERE activity_date BETWEEN '2019-06-28' AND '2019-07-27'
GROUP BY user_id) a
```



1148. Article Views I | Easy | [LeetCode](#)

Table: Views

TEXT

+-----+-----+		
Column Name	Type	
+-----+-----+		
article_id	int	
author_id	int	
viewer_id	int	
view_date	date	
+-----+-----+		

There is no primary key for this table, it may have duplicate rows.
Each row of this table indicates that some viewer viewed an article (written by s
Note that equal author_id and viewer_id indicate the same person.

Write an SQL query to find all the authors that viewed at least one of their own articles, sorted in ascending order by their id.

The query result format is in the following example:

TEXT

Views table:

+-----+-----+-----+-----+				
article_id	author_id	viewer_id	view_date	
+-----+-----+-----+-----+				
1	3	5	2019-08-01	
1	3	6	2019-08-02	
2	7	7	2019-08-01	
2	7	6	2019-08-02	
4	7	1	2019-07-22	
3	4	4	2019-07-21	
3	4	4	2019-07-21	
+-----+-----+-----+-----+				

Result table:

+-----+	
---------	--

```

| id    |
+-----+
| 4     |
| 7     |
+-----+

```

Solution

SQL

```

SELECT DISTINCT author_id AS id
FROM Views
WHERE author_id = viewer_id
ORDER BY author_id

```



1149. Article Views II | Medium | [LeetCode](#)

Table: views

TEXT

```

+-----+-----+
| Column Name | Type |
+-----+-----+
| article_id   | int  |
| author_id    | int  |
| viewer_id    | int  |
| view_date    | date |
+-----+-----+

```

There is no primary key for this table, it may have duplicate rows.

Each row of this table indicates that some viewer viewed an article (written by some author) on some date. Note that equal author_id and viewer_id indicate the same person.

Note that equal author_id and viewer_id indicate the same person.

Write an SQL query to find all the people who viewed more than one article on the same date, sorted in ascending order by their id.

The query result format is in the following example:

TEXT

Views table:

article_id	author_id	viewer_id	view_date
1	3	5	2019-08-01
3	4	5	2019-08-01
1	3	6	2019-08-02
2	7	7	2019-08-01
2	7	6	2019-08-02
4	7	1	2019-07-22
3	4	4	2019-07-21
3	4	4	2019-07-21

Result table:

id
5
6

Solution

SQL

```
SELECT DISTINCT viewer_id AS id#, COUNT(DISTINCT article_id) AS total
FROM views
GROUP BY viewer_id, view_date
HAVING count(DISTINCT article_id)>1
ORDER BY 1
```



Table: Users

TEXT

+-----+		
Column Name	Type	
+-----+		
user_id	int	
join_date	date	
favorite_brand	varchar	
+-----+		

user_id is the primary key of this table.

This table has the info of the users of an online shopping website where users ca

Table: Orders

TEXT

+-----+		
Column Name	Type	
+-----+		
order_id	int	
order_date	date	
item_id	int	
buyer_id	int	
seller_id	int	
+-----+		

order_id is the primary key of this table.

item_id is a foreign key to the Items table.

buyer_id and seller_id are foreign keys to the Users table.

Table: Items

TEXT

+-----+		
Column Name	Type	
+-----+		
item_id	int	
item_brand	varchar	

+-----+-----+

item_id is the primary key of this table.

Write an SQL query to find for each user, the join date and the number of orders they made as a buyer in 2019.

The query result format is in the following example:

TEXT

Users table:

+-----+-----+-----+			
user_id	join_date	favorite_brand	
+-----+-----+-----+			
1	2018-01-01	Lenovo	
2	2018-02-09	Samsung	
3	2018-01-19	LG	
4	2018-05-21	HP	
+-----+-----+-----+			

Orders table:

+-----+-----+-----+-----+-----+					
order_id	order_date	item_id	buyer_id	seller_id	
+-----+-----+-----+-----+-----+					
1	2019-08-01	4	1	2	
2	2018-08-02	2	1	3	
3	2019-08-03	3	2	3	
4	2018-08-04	1	4	2	
5	2018-08-04	1	3	4	
6	2019-08-05	2	2	4	
+-----+-----+-----+-----+-----+					

Items table:

+-----+-----+		
item_id	item_brand	
+-----+-----+		
1	Samsung	
2	Lenovo	
3	LG	
4	HP	

+-----+-----+

Result table:

+-----+-----+-----+		
buyer_id	join_date	orders_in_2019
+-----+-----+-----+		
1	2018-01-01	1
2	2018-02-09	2
3	2018-01-19	0
4	2018-05-21	0
+-----+-----+-----+		

Solution

SQL

```
SELECT user_id AS buyer_id, join_date, coalesce(a.orders_in_2019,0)
FROM users
LEFT JOIN
(
SELECT buyer_id, coalesce(count(*), 0) AS orders_in_2019
FROM orders o
JOIN users u
ON u.user_id = o.buyer_id
WHERE extract('year' FROM order_date) = 2019
GROUP BY buyer_id) a
ON users.user_id = a.buyer_id
```



1159. Market Analysis II | Hard | [LeetCode](#)

Table: Users

TEXT

+-----+-----+		
Column Name	Type	
+-----+-----+		
user_id	int	

join_date	date
favorite_brand	varchar

+-----+-----+

user_id is the primary key of this table.

This table has the info of the users of an online shopping website where users ca

Table: orders

TEXT

+-----+-----+

Column Name	Type
-------------	------

+-----+-----+

order_id	int
----------	-----

order_date	date
------------	------

item_id	int
---------	-----

buyer_id	int
----------	-----

seller_id	int
-----------	-----

+-----+-----+

order_id is the primary key of this table.

item_id is a foreign key to the Items table.

buyer_id and seller_id are foreign keys to the Users table.

Table: Items

TEXT

+-----+-----+

Column Name	Type
-------------	------

+-----+-----+

item_id	int
---------	-----

item_brand	varchar
------------	---------

+-----+-----+

item_id is the primary key of this table.

Write an SQL query to find for each user, whether the brand of the second item (by date) they sold is their favorite brand. If a user sold less than two items, report the answer for that user as no.

It is guaranteed that no seller sold more than one item on a day.

The query result format is in the following example:

TEXT

Users table:

user_id	join_date	favorite_brand
1	2019-01-01	Lenovo
2	2019-02-09	Samsung
3	2019-01-19	LG
4	2019-05-21	HP

Orders table:

order_id	order_date	item_id	buyer_id	seller_id
1	2019-08-01	4	1	2
2	2019-08-02	2	1	3
3	2019-08-03	3	2	3
4	2019-08-04	1	4	2
5	2019-08-04	1	3	4
6	2019-08-05	2	2	4

Items table:

item_id	item_brand
1	Samsung
2	Lenovo
3	LG
4	HP

Result table:

seller_id	2nd_item_fav_brand
-----------	--------------------

1	no
2	yes
3	yes
4	no

The answer for the user with id 1 is no because they sold nothing.

The answer for the users with id 2 and 3 is yes because the brands of their second

The answer for the user with id 4 is no because the brand of their second sold it

Solution

SQL



#Solution- 1:

```
SELECT user_id AS seller_id,
       IF(ISNULL(item_brand), "no", "yes") AS 2nd_item_fav_brand
FROM Users LEFT JOIN
(SELECT seller_id, item_brand
FROM Orders INNER JOIN Items
ON Orders.item_id = Items.item_id
WHERE (seller_id, order_date) IN
(SELECT seller_id, MIN(order_date) AS order_date
FROM Orders
WHERE (seller_id, order_date) NOT IN
(SELECT seller_id, MIN(order_date) FROM Orders GROUP BY seller_id)
GROUP BY seller_id)
) AS t
ON Users.user_id = t.seller_id and favorite_brand = item_brand
```

#Solution- 2:

```
WITH t1 AS(
SELECT user_id,
CASE WHEN favorite_brand = item_brand THEN "yes"
ELSE "no"
END AS 2nd_item_fav_brand
FROM users u LEFT JOIN
```

```
(SELECT o.item_id, seller_id, item_brand, RANK() OVER(PARTITION BY seller_id ORDE
FROM orders o join items i
USING (item_id)) a
ON u.user_id = a.seller_id
WHERE a.rk = 2)
```

```
SELECT u.user_id AS seller_id, COALESCE(2nd_item_fav_brand,"no") AS 2nd_item_fav_
FROM users u LEFT JOIN t1
USING(user_id)
```

1164. Product Price at a Given Date | Medium | [LeetCode](#)

Table: Products

TEXT

```
+-----+-----+
| Column Name | Type |
+-----+-----+
| product_id  | int  |
| new_price   | int  |
| change_date | date |
+-----+-----+
```

(product_id, change_date) is the primary key of this table.

Each row of this table indicates that the price of some product was changed to a

Write an SQL query to find the prices of all products on 2019-08-16. Assume the price of all products before any change is 10.

The query result format is in the following example:

TEXT

Products table:

```
+-----+-----+-----+
| product_id | new_price | change_date |
+-----+-----+-----+
| 1          | 20        | 2019-08-14  |
```

2	50	2019-08-14	
1	30	2019-08-15	
1	35	2019-08-16	
2	65	2019-08-17	
3	20	2019-08-18	
+-----+			

Result table:

+-----+		
product_id	price	
+-----+		
2	50	
1	35	
3	10	
+-----+		

Solution

SQL

#Solution- 1:

```
WITH t1 AS (
  SELECT a.product_id, new_price
  FROM(
    SELECT product_id, max(change_date) AS date
    FROM products
    WHERE change_date<='2019-08-16'
    GROUP BY product_id) a
  JOIN products p
  ON a.product_id = p.product_id AND a.date = p.change_date),

t2 AS (
  SELECT distinct product_id
    FROM products)

SELECT t2.product_id, coalesce(new_price,10) AS price
FROM t2 LEFT JOIN t1
ON t2.product_id = t1.product_id
ORDER BY price DESC
```



#Solution- 2:

```
SELECT t1.product_id AS product_id, IF(ISNULL(t2.price), 10, t2.price) AS price
FROM
  (SELECT distinct product_id
   FROM Products) AS t1 LEFT JOIN
  (SELECT product_id, new_price AS price
   FROM Products
   WHERE (product_id, change_date) in
        (SELECT product_id, max(change_date)
         FROM Products
         WHERE change_date <='2019-08-16'
         GROUP BY product_id)) AS t2
ON t1.product_id = t2.product_id
```

1173. Immediate Food Delivery I | Easy | [LeetCode](#)

Table: Delivery

TEXT

+-----+		
Column Name	Type	
+-----+		
delivery_id	int	
customer_id	int	
order_date	date	
customer_pref_delivery_date	date	
+-----+		

delivery_id is the primary key of this table.

The table holds information about food delivery to customers that make orders at

If the preferred delivery date of the customer is the same as the order date then the order is called immediate otherwise it's called scheduled.

Write an SQL query to find the percentage of immediate orders in the table, **rounded to 2 decimal places**.

The query result format is in the following example:

TEXT

Delivery table:

delivery_id	customer_id	order_date	customer_pref_delivery_date
1	1	2019-08-01	2019-08-02
2	5	2019-08-02	2019-08-02
3	1	2019-08-11	2019-08-11
4	3	2019-08-24	2019-08-26
5	4	2019-08-21	2019-08-22
6	2	2019-08-11	2019-08-13

Result table:

immediate_percentage
33.33

The orders with delivery id 2 and 3 are immediate while the others are scheduled.



Solution

SQL

#Solution- 1:

SELECT

ROUND(SUM(CASE WHEN order_date=customer_pref_delivery_date THEN 1 ELSE 0 END)/count(*)
FROM Delivery;

#Solution- 2:

SELECT

ROUND(avg(CASE WHEN order_date=customer_pref_delivery_date THEN 1 ELSE 0 END)*100
FROM delivery

